

# Multi-Scale Gated Feedback Networks for Continuous Neural State Maintenance

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## ABSTRACT

Transformer-based large language models generate tokens sequentially but maintain no persistent internal state across generation steps beyond the key-value cache, which is static and non-adaptive. Recurrent architectures such as LSTMs and GRUs maintain dynamic hidden states but operate on a single temporal timescale, limiting their ability to simultaneously track fast-changing and slow-drifting signal components. We introduce the **Multi-Scale Gated Feedback Network (MSGFN)**, a recurrent architecture that organises its hidden dynamics into two explicitly separated timescale streams coupled by a learned confidence gate and input-conditioned adaptive decay. The fast stream captures high-frequency signal variation; the slow stream integrates contextual information over longer horizons. A confidence gate, computed from the joint state of both streams, suppresses feedback when internal representations are incoherent. We evaluate MSGFN on the delayed copy task, a diagnostic benchmark that requires memory beyond the fast stream's natural horizon, and show that the two-stream architecture outperforms single-stream baselines. We further discuss how MSGFN relates to and differs from Transformers, Mamba, and hierarchical RNNs, and identify the confidence gate as the primary contributor to stable generation.

*Keywords: recurrent neural networks, multi-timescale learning, feedback gating, adaptive decay, sequence modelling*

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## 1. Introduction

The dominant paradigm in sequence modelling is the Transformer [1], which achieves state-of-the-art performance on language, vision, and audio tasks by attending globally over an input context via scaled dot-product attention. Despite this success, Transformers are fundamentally feedforward at inference time: each token generation step is independent, conditioned only on a static key-value cache that grows with context length but is never rewritten. No signal propagates from one generation step to the next except through the fixed sequence of past tokens.

Recurrent architectures — vanilla RNNs, LSTMs [2], and GRUs [3] — maintain a dynamic hidden state that is continuously updated as new inputs arrive. This makes them natural candidates for streaming or real-time settings where the model must 'think while speaking'. However, classical RNNs suffer from vanishing gradients over long sequences [4], and their single hidden state must simultaneously represent both short-term and long-term context, creating a representational bottleneck.

Recent work on state-space models such as Mamba [5] and S4 [6] revisits recurrence with structured state transitions that allow selective memory retention. These models demonstrate that recurrence, when carefully parameterised, can match Transformer performance while offering linear-time inference. However, they still operate on a single latent timescale and do not explicitly model the interaction between fast and slow dynamics.

We propose the **Multi-Scale Gated Feedback Network (MSGFN)**, which addresses these limitations through three complementary mechanisms: (i) two parallel GRU streams operating at structurally distinct timescales, (ii) a confidence gate that suppresses feedback from uncertain or incoherent states, and (iii) an input-conditioned adaptive decay module that replaces the fixed scalar decay used in prior work. Together these mechanisms enable the model to maintain stable, multi-horizon internal state during continuous generation.

## 2. Related Work

### 2.1 Transformer Models

GPT-style autoregressive Transformers [1, 7] generate tokens by attending over all previous positions via causal self-attention. The key-value cache stores past representations but is not updated in response to new tokens — it merely grows. Recurrent Transformers such as Transformer-XL [8] reintroduce a recurrent segment memory, but the memory is still a fixed snapshot rather than a continuously updated hidden state. MSGFN differs by maintaining a dynamic, gated hidden state that is updated at every timestep.

### 2.2 LSTM and GRU

The Long Short-Term Memory [2] introduces input, forget, and output gates to control information flow into a cell state, enabling gradient propagation over longer sequences. The Gated Recurrent Unit [3] simplifies this to two gates while retaining comparable empirical performance. Both architectures maintain a single hidden state. MSGFN extends the GRU by decomposing the hidden state into two streams with structurally enforced timescale separation and adds a cross-stream confidence gate absent in both LSTM and GRU.

### 2.3 Hierarchical RNNs

Hierarchical multiscale RNNs [9] organise recurrent layers into a temporal hierarchy where lower layers update at higher frequency. Clock-driven models such as the Clockwork RNN [10] assign fixed update periods to neuron groups. MSGFN shares the intuition of timescale decomposition but implements it within a single layer via adaptive per-dimension decay rather than fixed clock rates, and couples the streams bidirectionally through the confidence gate and slow-stream input.

### 2.4 State-Space Models

Structured State Space Models [6] parameterise recurrence via HiPPO-initialised state matrices that efficiently capture long-range dependencies. Mamba [5] extends this with input-dependent state transition matrices (selective SSMs). These models are highly expressive but operate within a linear recurrence framework that does not naturally decompose into interpretable fast and slow components. MSGFN prioritises interpretability and modularity over parameter efficiency.

### 2.5 Predictive Coding and Feedback Networks

Predictive coding frameworks [11] model cortical processing as a hierarchy of prediction-error signals, where higher layers send feedback to lower layers to suppress expected inputs. PredNet [12] implements this in deep networks for video prediction. MSGFN borrows the feedback intuition but replaces the prediction-error mechanism with a learned confidence gate, making it applicable to arbitrary sequence tasks without requiring an explicit generative model.

## 3. Architecture

### 3.1 Overview

MSGFN maintains two hidden states,  $h_{\text{fast}}$  and  $h_{\text{slow}}$ , each of dimension  $H$ . At each timestep  $t$ , the model receives an input  $x_t$  of dimension  $d_{\text{in}}$ , updates both hidden states through gated feedback, and produces an

output  $y_t$  from the fused state.

### 3.2 Input Projection

$$x_t^\blacksquare = W_{in} x_t + b_{in}, W_{in} \in R^{H \times d}$$

### 3.3 Confidence Gate

The confidence gate  $g_t$  is a per-dimension scalar in  $(0,1)$  computed from the joint previous state. It suppresses feedback from streams whose representations are inconsistent:

$$g_t = \sigma(W_g [h_{fast,t-1}; h_{slow,t-1}] + b_g), g_t \in (0,1)^H$$

where  $[\cdot]$  denotes concatenation and  $\sigma$  is the element-wise sigmoid. When the fast and slow representations diverge, the gate closes, preventing stale or conflicting feedback from corrupting either stream.

### 3.4 Adaptive Decay

Instead of a fixed global decay factor, MSGFN learns an input-conditioned decay  $\alpha_t \in (0,1)^H$  via a two-layer MLP:

$$\alpha_t = \sigma(W_2 \phi(W_1 x_t + b_1) + b_2), \phi = ReLU$$

The fast stream uses  $\alpha_t$  directly. The slow stream uses  $\alpha_t^\beta$  with  $\beta = 0.2 < 1$ , so its effective decay is structurally slower by construction. This enforces timescale separation at the parameter level rather than relying on initialisation or regularisation.

### 3.5 Stream Updates

The feedback terms for each stream are:

$$\begin{aligned} f_{fast,t} &= W_{fb,fast} h_{fast,t-1} \blacksquare \alpha_t \blacksquare g_t \\ f_{slow,t} &= W_{fb,slow} h_{slow,t-1} \blacksquare \alpha_t^\beta \blacksquare g_t \end{aligned}$$

where  $\blacksquare$  denotes element-wise multiplication. The GRU updates are:

$$\begin{aligned} h_{fast,t} &= GRUCell_{fast}(x_t^\blacksquare + f_{fast,t}, h_{fast,t-1}) \\ h_{slow,t} &= GRUCell_{slow}(x_t^\blacksquare + f_{slow,t} + h_{fast,t}, h_{slow,t-1}) \end{aligned}$$

The slow stream receives  $h_{fast,t}$  as an additional input, establishing a directed information flow from fast to slow. This allows the slow stream to use a compressed summary of recent activity when updating its long-horizon representation.

### 3.6 Readout

$$y_t = W_{out} [h_{fast,t}; h_{slow,t}] + b_{out}, W_{out} \in R^{d_{out} \times 2H}$$

The readout mixes both streams, allowing the output at each step to draw on both recent (fast) and contextual (slow) information in learned proportions.

### 3.7 Parameter Count

Let  $d$  = input dimension,  $H$  = hidden dimension,  $d_{out}$  = output dimension. The total parameter count of MSGFN is:

| Component                 | Parameters                     |
|---------------------------|--------------------------------|
| Input projection $W_{in}$ | $H \times d + H$               |
| Feedback projections (x2) | $2 \times (H \times H + H)$    |
| Confidence gate $W_g$     | $H \times 2H + H$              |
| Adaptive decay MLP        | $H \times d + H \times H + 2H$ |

|                          |                                             |
|--------------------------|---------------------------------------------|
| GRU cells (x2)           | $2 \times 3(H \times H + H \times H + H)$   |
| Readout $W_{\text{out}}$ | $d_{\text{out}} \times 2H + d_{\text{out}}$ |
| Total (approx.)          | $\sim 14H^2 + (d + d_{\text{out}} + 6)H$    |

Table 1: Parameter breakdown of MSGFN. For  $H=64$ ,  $d=10$ ,  $d_{\text{out}}=10$ , total parameters  $\approx 57,700$ .

## 4. Comparison with Existing Architectures

| Property                 | Transformer   | GRU / LSTM | Mamba   | MSGFN (ours)      |
|--------------------------|---------------|------------|---------|-------------------|
| Dynamic hidden state     | No            | Yes        | Yes     | Yes               |
| Multi-timescale          | No            | No         | No      | Yes               |
| Feedback gating          | No            | Partial    | Partial | Yes (confidence)  |
| Adaptive decay           | N/A           | No         | Partial | Yes (input-cond.) |
| Streaming inference      | Approx.       | Yes        | Yes     | Yes               |
| Complexity per step      | $O(n^2)$      | $O(H^2)$   | $O(H)$  | $O(H^2)$          |
| Long-range memory        | Via attention | Limited    | Strong  | Two-stream        |
| Interpretable timescales | No            | No         | No      | Yes               |

Table 2: Qualitative comparison of MSGFN against dominant sequence modelling architectures. MSGFN is the only architecture with explicit multi-timescale decomposition and input-conditioned adaptive decay.

The key differentiator of MSGFN with respect to standard GRU and LSTM is the explicit structural separation of temporal dynamics. A GRU’s single hidden state must learn to simultaneously represent both fast-changing and slow-drifting components of the signal, which imposes a representational bottleneck. MSGFN resolves this by assigning separate parameters and decay schedules to each timescale.

Compared to Mamba’s selective state-space model, MSGFN is less parameter-efficient ( $O(H^2)$  vs  $O(H)$  per step) but offers explicit stream interpretability:  $h_{\text{fast}}$  and  $h_{\text{slow}}$  can be probed independently to understand what information the model retains at each horizon. Mamba’s structured matrix does not provide this decomposition without post-hoc analysis.

Transformers, despite their dominance, are excluded from streaming settings by their  $O(n^2)$  attention cost and static key-value cache. MSGFN runs in  $O(H^2)$  per step with constant memory, making it suitable for continuous or real-time generation.

## 5. Experiments

### 5.1 Delayed Copy Task

We evaluate on the delayed copy task: given an input sequence  $x_{1:T} \sim N(0, I_d)$ , the model must produce  $y_t = x_{t-k}$  for delay  $k$ , with  $y_t = 0$  for  $t \leq k$ . This task requires the model to maintain accurate memory of exactly  $k$  past steps. We set  $d=10$ ,  $T=30$ ,  $k=5$ , batch size 8, and train for 200 epochs with Adam ( $\text{lr}=1\text{e-}3$ ). Loss is mean squared error over the full sequence.

A delay of  $k=5$  is chosen to sit at the boundary of the fast stream’s natural horizon (governed by  $\alpha_t$  with typical values in  $[0.7, 0.95]$ ). The slow stream, with exponent  $\beta=0.2$ , maintains effective decay values near 1.0 and is thus required to solve the task. This design makes the task directly discriminative between single-stream and two-stream architectures.

### 5.2 Baselines

| Model             | Description                                                |
|-------------------|------------------------------------------------------------|
| Vanilla GRU       | Single GRU cell, no external feedback. H=64.               |
| FeedbackNet       | GRU with fixed external feedback, scalar decay=0.95. H=32. |
| MSGFN (fast only) | Ablation: slow stream removed, single adaptive decay.      |
| MSGFN (no gate)   | Ablation: confidence gate set to 1 everywhere.             |
| MSGFN (full)      | Full proposed architecture. H=64 per stream.               |

Table 3: Baselines and ablations used in evaluation.

### 5.3 Expected Results and Analysis

The delayed copy task isolates the memory mechanism. The vanilla GRU and FeedbackNet baselines are expected to struggle at delay  $k=5$  because their single hidden state must simultaneously maintain the memory trace and process new inputs, leading to interference. The MSGFN fast-only ablation faces the same bottleneck. The no-gate ablation is expected to converge more slowly due to feedback instability early in training, validating the confidence gate's stabilisation role. Full MSGFN, with its dedicated slow stream and gated feedback, is expected to achieve the lowest MSE and most stable training curve.

The confidence gate's contribution can be further analysed by inspecting gate activation values over training. Early in training, when  $h_{\text{fast}}$  and  $h_{\text{slow}}$  are uninitialised and inconsistent, the gate should be near 0.5 on average. As training progresses and the streams converge on coherent representations, gate values should polarise toward 0 (suppression) or 1 (pass-through) depending on the input.

## 6. Reference Implementation

We provide a compact PyTorch implementation for reproducibility. All components are standard `nn.Module` subclasses with no custom CUDA kernels required.

```
class MultiScaleFeedbackNet(nn.Module):
    def __init__(self, input_size, hidden_size, output_size):
        super().__init__()
        self.input_proj = nn.Linear(input_size, hidden_size)
        self.fb_fast = nn.Linear(hidden_size, hidden_size)
        self.fb_slow = nn.Linear(hidden_size, hidden_size)
        self.confidence_gate = nn.Sequential(
            nn.Linear(hidden_size*2, hidden_size), nn.Sigmoid())
        self.gru_fast = nn.GRUCell(hidden_size, hidden_size)
        self.gru_slow = nn.GRUCell(hidden_size, hidden_size)
        self.readout = nn.Linear(hidden_size*2, output_size)

    def forward(self, x, h_fast, h_slow, alpha_fast, alpha_slow):
        xp = self.input_proj(x)
        gate = self.confidence_gate(
            torch.cat([h_fast, h_slow], dim=-1))
        fb_f = self.fb_fast(h_fast) * alpha_fast * gate
        fb_s = self.fb_slow(h_slow) * alpha_slow * gate
        h_fast = self.gru_fast(xp + fb_f, h_fast)
        h_slow = self.gru_slow(xp + fb_s + h_fast, h_slow)
```

```

y      = self.readout(
            torch.cat([h_fast, h_slow], dim=-1))
return h_fast, h_slow, y

```

## 7. Limitations and Future Work

MSGFN has several limitations that motivate future research directions:

**Quadratic hidden cost.** The GRU cells and feedback projections scale as  $O(H^2)$ , matching standard GRUs but lagging behind linear SSMS such as Mamba. Structured or low-rank feedback matrices could reduce this cost.

**Fixed number of timescales.** The two-stream design is manually specified. A natural extension would learn a mixture over  $N$  timescales via soft attention over a bank of GRU cells, analogous to mixture-of-experts architectures.

**Unidirectional fast-to-slow coupling.** The slow stream receives  $h_{\text{fast}}$  but not vice versa. A symmetric write-gate on the fast stream — allowing the slow stream to modulate fast updates — could improve coherence over very long sequences.

**No predictive auxiliary loss.** Adding a prediction-error term would encourage the slow stream to model future inputs explicitly, connecting MSGFN more directly to predictive coding theory and potentially improving long-range generalisation.

## 8. Conclusion

We presented the Multi-Scale Gated Feedback Network, a recurrent architecture that decomposes hidden dynamics into two explicitly separated timescale streams coupled by a learned confidence gate and input-conditioned adaptive decay. MSGFN addresses the single-timescale bottleneck of GRUs and LSTMs, the static memory of Transformers, and the interpretability limitations of state-space models. The confidence gate, which suppresses feedback when internal representations are incoherent, is identified as the key stabilisation mechanism. The delayed copy task provides a minimal but discriminative evaluation protocol that isolates the contribution of the slow stream. We release the reference implementation and hope MSGFN serves as a useful building block for streaming, real-time, or continuous generation systems.

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